Exploratory Data Analysis (EDA) Summary Report

1. Introduction

This report provides an exploratory data analysis (EDA) of Bike sales to assess data quality, identify key insights, and highlight factors that influence decision making. The goal of this analysis is to ensure data readiness for predictive modeling and risk assessment.

2. Dataset Overview

The dataset contains 1026 records, representing Bike sales with key attributes relevant to information of customers. It includes numerical and categorical features such as Id, Income, Region, Commute Distance, Age etc.

Key dataset attributes:

- Number of records: 1026
- Key variables: Age, Income, Purchased Bike, Region, Cars, House owner, Gender
- Data types: Categorical (Martial status, Gender, Education, Occupation, House Owner, Commute Distance, Region, Purchased Bike), Numerical (Income, Id, Children, Age, Cars)

3. Duplicates in Data Analysis

Some data of the customers was noted-down twice in the raw data. These duplicates are removed and I have worked on the cleaned dataset. And there are no null values in the data.

Key data findings:

- Total number duplicated rows are 26.
- After dropping those 26 duplicated rows, I got cleaned data with 1000 rows and 13 columns

4. Key Observations

- The average income of the customers in the data is nearly equal to 60000
- Non-customers who haven't purchased the bike has the income distribution between 40000 and 60000
- Comparatively there are more number of customers who owns the car and not purchased the bike
- Majority of customers irrespective of bike purchase are from North-America region(50.8%)

- There is no big difference between the male and female customers.
- There are customers who purchased bike have own houses (67.6%)
- Mean of 44 aged people are the customers in our sales
- Though North America region has more customers percentage, but we can see bike purchase percentage high in Pacific region
- Most of the customers in North America are non-buyers who owns a house and cars and most of the customers in Pacific region are buyers who owns a car and house.
- Average income and average age of the all non-buyers irrespective of regions are around 60000 Dollars and 47 years

5. Conclusion & Next Steps

This EDA analysis paints a picture of a **successful, mass-market product targeted at established, middle-class families in North America.** The typical bike buyer is not an extreme athlete or a low-income individual reliant on a bike for transport, but rather a homeowner with a comfortable income, likely purchasing for leisure, fitness, or family activities.

Key Findings:

1. **Target Audience Confirmed:** The ideal customer is **middle-aged (~44 years old), middle-income (~$60K), and a homeowner**. This suggests financial stability and a lifestyle where a bike is a discretionary purchase for enjoyment.
2. **The North American Home Advantage:** The overwhelming majority of market is in **North America**. This isn't just largest market; it's dominant one, representing over half of all potential customers.
3. **The Car Owner Paradox:** The fact that most customers (and non-customers) own cars confirms that the bike is not their primary transportation. It's a **secondary, leisure product**. The challenge is converting car-owning non-customers.
4. **Gender Neutrality:** The product appeal is **equal across genders**, which simplifies marketing messaging—it doesn't need to be gendered.

Next Steps:

- **Double Down on North America:** Marketing resources and budget should be heavily focused here. This is core, proven market. Consider regional campaigns within North America (e.g., "Summer biking in California," "Trail biking in Colorado").
- **Reframe the Messaging:** Target **homeowners and car owners**. Messaging should focus on **leisure, family fun, health, and convenience** (e.g., "The perfect weekend activity for your family," "Skip the traffic, enjoy a ride in local park," "A great way to stay fit without the gym").
- **Leverage the Homeowner Insight:** Partner with home-related brands or advertise in publications/websites targeting homeowners.